

Loading Common Voice Arabic dataset (streaming) ...

✓ Streaming dataset ready

Extracting features from 50 utterances ...

10/50 utterances – 35s ...

20/50 utterances – 53s ...

30/50 utterances – 72s ...

40/50 utterances – 103s ...

50/50 utterances – 123s ...

Done in 122.9s

Class distribution: {'D': 76, 'H': 356, 'S': 205, 'T': 205, 'Z': 60, 'aa': 1222, 'ain': 470, 'b': 603, 'd': 377, 'dh': 182, 'f': 233, 'gh': 46, 'h': 538, 'hamza': 956, 'k': 1000, 'l': 1000, 'm': 1000, 'n': 1000, 'o': 1000, 'p': 1000, 'q': 1000, 'r': 1000, 'sh': 1000, 'ss': 1000, 't': 1000, 'th': 1000, 'v': 1000, 'w': 1000, 'y': 1000, 'z': 1000}

✓ Feature matrix : (14766, 90)

✓ Unique phonemes : 29 → [np.str_('D'), np.str_('H'), np.str_('S'), np.str_('T'), np.str_('Z'), np.str_('aa'), np.str_('ain'), np.str_('b'), np.str_('d'), np.str_('dh'), np.str_('f'), np.str_('gh'), np.str_('h'), np.str_('hamza'), np.str_('k'), np.str_('l'), np.str_('m'), np.str_('n'), np.str_('o'), np.str_('p'), np.str_('q'), np.str_('r'), np.str_('sh'), np.str_('ss'), np.str_('t'), np.str_('th'), np.str_('v'), np.str_('w'), np.str_('y'), np.str_('z')]

Fitting RF (300 trees, 90-dim) ...

✓ Fit done in 37.3s

✓ OOB accuracy : 0.3678

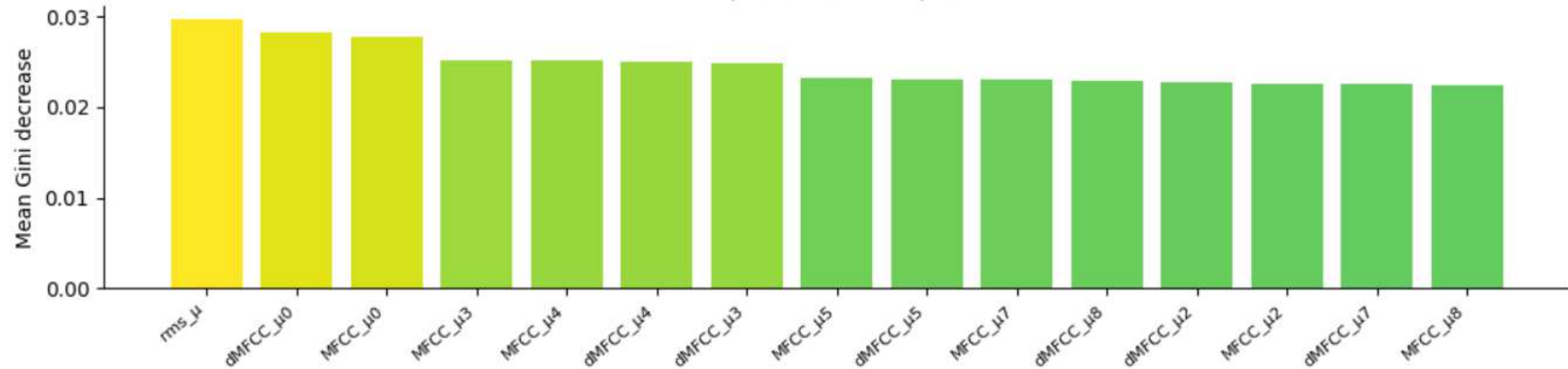
✓ Test accuracy : 0.4185

✓ Classes learned: [np.str_('D'), np.str_('H'), np.str_('S'), np.str_('T'), np.str_('Z'), np.str_('aa'), np.str_('ain'), np.str_('b'), np.str_('d'), np.str_('dh'), np.str_('f'), np.str_('gh'), np.str_('h'), np.str_('hamza'), np.str_('k'), np.str_('l'), np.str_('m'), np.str_('n'), np.str_('o'), np.str_('p'), np.str_('q'), np.str_('r'), np.str_('sh'), np.str_('ss'), np.str_('t'), np.str_('th'), np.str_('v'), np.str_('w'), np.str_('y'), np.str_('z')]

RF — Top-15 feature importances

0.03 ↓

RF — Top-15 feature importances



 Reference sentence : فكانت هباء منبثا

↳ Change EXPECTED_TEXT before uploading your audio.

 Upload your Arabic audio file (wav / mp3 / m4a)

 RF phoneme-set matching | ASR = WER reference only

Sélect. fichiers Enregistrement.m4a

Enregistrement.m4a(audio/x-m4a) - 98028 bytes, last modified: 28/04/2026 - 100% done

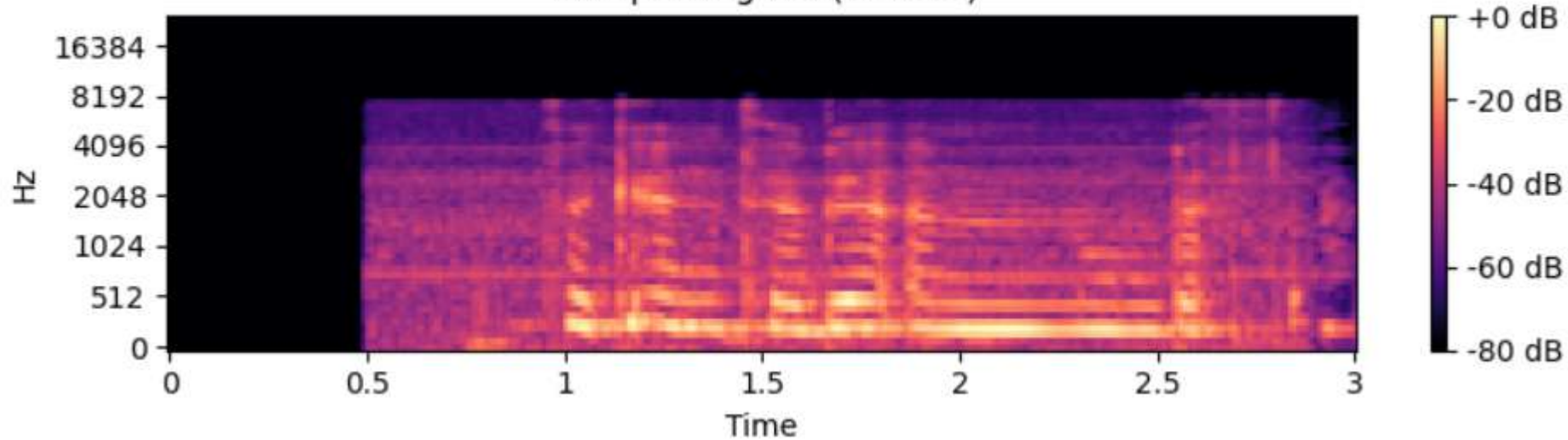
Saving Enregistrement.m4a to Enregistrement.m4a

 Processing : Enregistrement.m4a

Duration : 3.82s | SR: 48000 Hz

— STEP 1 • Mel-spectrogram —

Mel-spectrogram (first 3 s)



— STEP 2 • G2P (reference text) —

Text : فكاكت هباء منبثا

Words : ['فكاكت', 'هباء', 'منبثا']

فكاكت → ['f', 'k', 'aa', 'n', 't']

هباء → ['h', 'b', 'aa', 'hamza']

منبثا → ['m', 'n', 'b', 'th', 'aa']

— STEP 3 • VAD + per-word RF prediction —

فكاكت → ['aa', 'n', 'l', 'silence', 'n', 'aa', 'm', 'aa', 'n', 'm', 'silence', 'l', 'n', 'l', 'n', 'silence', 'k', 'y', 't', 'hamza', 'r', 'silence', 'y', 'silence']

هباء → ['k', 'n', 'silence', 'l', 'silence', 'aa', 'silence', 'l', 'r', 'm', 'aa', 'silence', 'r', 'aa', 'silence', 'y', 'silence', 'aa', 'silence', 'n', 'silence']

منبثا → ['aa', 'silence', 'b', 'l', 'silence', 'l', 'silence', 'n', 'silence', 'aa', 'silence', 'l', 'y', 'silence', 'y', 'l', 'm', 'silence', 'y', 'n', 'y', 'l']

— STEP 4 • Phoneme-set matching —

— STEP 5 • Report & visualisations —

ARABIC PRONUNCIATION REPORT (RF phoneme-set analysis)

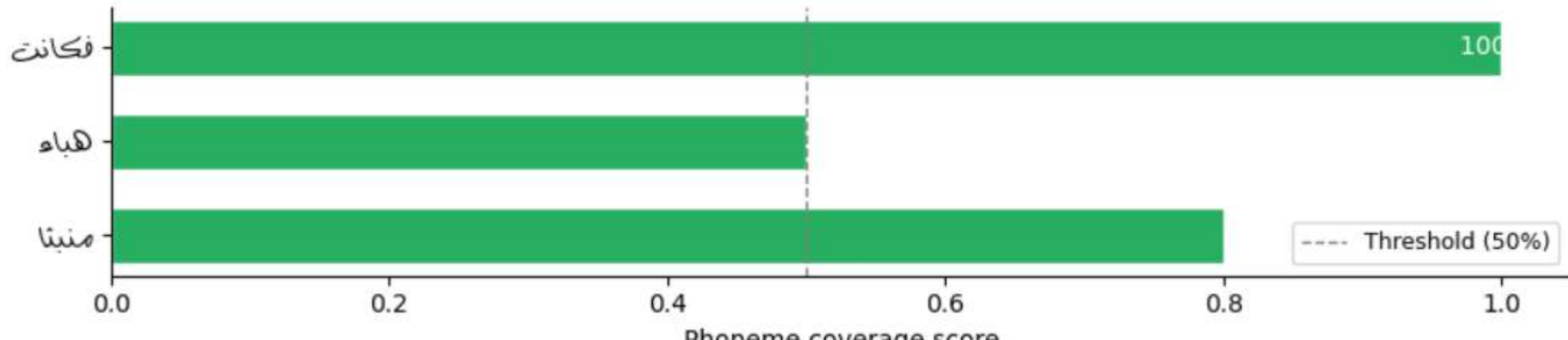
Match threshold : $\geq 50\%$ expected phonemes present

Words correct : 3 / 3

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- ✓ فُكَّانَت (coverage 100%) ⚠ low RF confidence
 - ✓ هَبَاء (coverage 50%) ⚠ low RF confidence
 - ✓ مَنبِئًا (coverage 80%) ⚠ low RF confidence
-

 Perfect pronunciation!

Pronunciation — RF phoneme coverage per word



Expected

f k aa n t

h b ah am

m n b th aa

RF detected

aa n silence aa m aa n silence n l silence k y hamz silence silence f silence hamz silence silence silence silence hamz

k silence silence silence r m silence silence silence silence l y silence silence silence silence silence r

aa silence b silence silence silence l y silence l n y silence silence l silence m r

— STEP 6 • ASR WER reference (informational) —

Loading ASR model (WER reference only) ...

Loading weights: 100%  424/424 [00:00<00:00, 637.19it/s, Materializing param=wav2vec2.masked_spec_embed]

ASR ready in 4.6s

ASR output : فكاكت هباء منبفة

WER : 33.33% ← reference only